CONSERVATION OF MASS

Name: ..... **Date:** .....

Part 1: count the atoms of each element on each side and write Yes or No in the last column. Part 2: use the law to find the missing masses. Part 3: answer in full sentences.

Law of conservation of mass: atoms are not created or destroyed in a chemical reaction, so the total mass of the products equals the total mass of the reactants.

PART 1: IS IT BALANCED?

Equation	Atoms on the left	Atoms on the right	Balanced?
$H_2 + Cl_2 \rightarrow 2HCl$			
$Mg + O_2 \rightarrow MgO$			
$2H_2 + O_2 \rightarrow 2H_2O$			
$N_2 + H_2 \rightarrow NH_3$			
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$			
$Zn + HCl \rightarrow ZnCl_2 + H_2$			

PART 2: FIND THE MISSING MASS

1. 24 g of magnesium burns completely in 16 g of oxygen. Mass of magnesium oxide formed: _____ g
2. 100 g of calcium carbonate decomposes into 56 g of calcium oxide and _____ g of carbon dioxide.
3. 4 g of hydrogen reacts with 32 g of oxygen to form _____ g of water.
4. 10 g of iron reacts with sulfur to make 15.7 g of iron sulfide. Mass of sulfur that reacted: _____ g

PART 3: EXPLAIN

When a log burns, the ash left behind weighs much less than the log. Does this break the law of conservation of mass? Explain.

CONSERVATION OF MASS ANSWER KEY

Name: ..... **Date:** .....

PART 1: IS IT BALANCED?

$H_2 + Cl_2 \rightarrow 2HCl$ | H 2, Cl 2 | H 2, Cl 2 | Yes

$Mg + O_2 \rightarrow MgO$ | Mg 1, O 2 | Mg 1, O 1 | No

$2H_2 + O_2 \rightarrow 2H_2O$ | H 4, O 2 | H 4, O 2 | Yes

$N_2 + H_2 \rightarrow NH_3$ | N 2, H 2 | N 1, H 3 | No

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ | C 1, H 4, O 4 | C 1, H 4, O 4 | Yes

$Zn + HCl \rightarrow ZnCl_2 + H_2$ | Zn 1, H 1, Cl 1 | Zn 1, Cl 2, H 2 | No

PART 2: FIND THE MISSING MASS

1. 40 g 2. 44 g 3. 36 g 4. 5.7 g

PART 3: EXPLAIN

1. No. The wood reacts with oxygen from the air and most of the products (carbon dioxide and water vapour) escape as gases. Ash + gases = wood + oxygen used, so mass is conserved.

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